

Candidate Task Brief

Title: AI-Driven Formula 1 Driver & Team Performance Tracker (MVP)

Duration: 16–18 hours

Deadline: Within 48 hours

Submission Format: GitHub repo + short walkthrough video (2–5 mins) + task report (PDF)

Task Objective

Design a compact, insightful AI-powered tool that analyzes and visualizes the performance of Formula 1 drivers and teams over time using publicly available datasets. Use AI agents (scripts or basic workflows) to detect trends and anomalies in the data and present your findings in a simple dashboard.

Core Deliverables

1. Dataset Integration

- Use an open Formula 1 dataset (e.g., [Kaggle F1 dataset](#) or Ergast Developer API).
- Clean, preprocess, and structure data into usable formats for analysis (e.g., lap times, pit stops, qualifying, race results).

2. Dashboard (Basic MVP using Streamlit/Jupyter/Gradio/etc.)

- Visual insights: Driver performance, team comparison, season progression.
- At least 4 meaningful graphs or tables (e.g., lap consistency, podium frequency, average grid start vs finish).

3. AI Agent 1: Performance Trend Analyzer

- Predict performance trends using linear regression or classification models.
- Example: Predict if a driver will improve or decline in next 3 races based on past races.

4. AI Agent 2: Anomaly Detector

- Detect significant deviations from normal performance patterns.
- Flag issues like unusual pit stop behavior, sudden lap time shifts, etc.

5. Alert System (Optional Bonus)

- Simulate an alert system (could be in-code or mock notification) to notify when anomaly is detected.

6. Documentation & Reporting

- Include:
 - Data pipeline description
 - ML model explanation
 - Key findings and insights
 - What you would improve in V2

Tech Stack Suggestions

- Languages: Python (mandatory)
- Libraries: Pandas, Seaborn/Plotly, Scikit-learn, Streamlit/Gradio/Plotly Dash
- Bonus: TensorFlow, OpenCV (only if useful in modeling), Ergast API

Evaluation Criteria

- Clarity of insights & visuals
- Code quality, structure, and documentation
- Creativity in anomaly and trend detection
- Simplicity and usability of the MVP
- Clear understanding of data pipeline and modeling logic

Submission Checklist

- GitHub Repo (well-structured, with README)
- Short video walkthrough (2–5 minutes)
- Report/Slide Deck/PDF with insights
- Optional: Demo Link (if hosted)

